

# Compiler-Validated Semantic Editing for Weak Local Coding Models

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## Abstract

Coding agents edit source code as text, and small locally runnable models fail at it: diff formats break, identifiers get guessed, and errors surface only after files change. ago is a semantic edit protocol for Go. The agent queries a typechecked workspace and submits mutations from a versioned operation catalog; the compiler validates every mutation against an in-memory snapshot before anything reaches disk. Edits apply as atomic, generation-checked transactions or not at all, so an edit that does not typecheck is unrepresentable, not merely checked. Rejections are the conversation: each carries typed diagnostics and paste-ready repairs (the corrected call, verbatim), and resending a rejected call unchanged escalates. An oracle replays each task’s ground-truth commit through the protocol before any model attempts it, so a task enters the benchmark only once it is proven solvable; tasks are mined from real commits in traefik, vault, and boundary. The evaluation is a three-arm grid, raw file editing, LSP symbol addressing with advisory validation (Serena), and the protocol, run against a frozen engine so the benchmark cannot chase the tasks. Across three locally served models the arm ordering never inverts: the protocol beats symbol addressing, which beats raw text editing. The floor 9B model passes 41% of episodes through the protocol against 32% with symbol addressing alone and 24% raw; the widest gap is the model that is a poor freeform tool-caller, whose raw pass rate collapses to 4% yet recovers to 53% through the protocol. Sampled on-disk states during editing fail to compile 76%–86% of the time in the raw arm and at most 2% in the semantic arm, in every model. All code, per-episode evidence (transcripts, request logs, serving configurations), and analysis tooling are released.

## 1 Introduction

TODO(guy): expand from idea.md.

## 2 Related Work

Symbol-level agent tooling over LSP (Serena; gopls’s experimental MCP server) navigates semantically but still edits text, or exposes a single mutation. The compiler diagnostics sit right there in those stacks; the difference here is that validation is not advisory. CODESTRUCT frames agents over structured action spaces with syntax-validated edits and reports token and accuracy gains; ago’s mutations are type-checked, transactional, and carry repairs. Editing-format benchmarks (EDIT-Bench, RES-Q, FEA-Bench) measure text formats and find that strict formats hurt current models. That is the failure mode the repair channel converts: the

format stays strict and the rejection hands back the corrected call. Compiler-guided patch generation (Repilot) and typed-hole context (Hazel) take the same validate-during-generation stance at expression granularity. TODO: full citations.

## 3 The ago Protocol

### 3.1 Surface

Ten tools carry a versioned catalog of 33 operations across declaration, statement, test, and project families, plus seven query kinds with deterministic, paged responses. Node handles address statements inside declarations; generation counters make staleness explicit instead of silent.

### 3.2 Transactions

An ordered op list validates as one unit against an in-memory snapshot. Ops apply to a copy, the dirty set re-typechecks once (in parallel, scheduled over post-edit import edges), resolution proofs run (rename rejects reference capture the compiler would happily accept), then all files write together or nothing does. Pre-existing issues in the affected packages are baselined and tolerated; real codebases carry history, and only new diagnostics reject.

### 3.3 Rejection as conversation

Every rejection answers one question: what is the correct next call. It carries typed diagnostics, candidate lists, and `possible_repairs`, complete calls the agent resends verbatim. Repairs are tested by execution before they are offered; a repair that would itself reject is worse than none. Resending a rejected call unchanged escalates. Accepted mutations embed the fresh view of the touched declaration, so back-to-back edits skip the read round trip.

## 4 Oracle Methodology

Every benchmark task comes from a real commit, so the correct answer is already known. Before any model attempts a task, the oracle submits that known answer through the protocol itself. If the right answer cannot get through, the model never had a chance: the problem is ours (an engine bug, an extractor error, or a named protocol ceiling), and the task stays out of the roster until it is fixed. This separates “the model failed” from “the task was impossible” by construction, and the oracle transcripts double as a training corpus.

## 5 Benchmark

Same model, same harness, same prompts (token parity is enforced by test), same time cap, three arms. Raw edits files with shell tools. Serena addresses symbols through the language server (a pinned revision of the Serena MCP server) but validation stays advisory: diagnostics are available on request and nothing blocks a broken write from landing. Semantic sees only the protocol. The middle arm is the ablation that separates the two variables the protocol bundles: symbol addressing and mandatory validation. Scoring is a goal predicate plus a clean typecheck plus scoped tests, and the test gate is voided where a pristine worktree also fails it.

Serving configurations are pinned per run: GLM-4.7-Flash (MoE) at UD-Q4\_K\_XL, roughly 20 GiB with a q8\_0 K/V cache at 128k context; gpt-oss-20b at MXFP4, 11.28 GiB of weights before KV; Qwen3.5-9B at UD-Q4\_K\_XL, its card’s embedding server evicted to fit the larger quant. Local serving is part of the experiment, constraints included. Every episode records its transcript, diff, score, token usage, failure kind, and the daemon’s request log; the evidence is the contribution as much as the numbers.

## 5.1 Evaluation freeze

Every run before the freeze tag is development data: rejections became repairs and oracle failures became engine fixes, a loop that is a contribution but cannot also be the headline evidence. The engine is frozen at a tagged revision (**eval-freeze-1**); every episode records the engine revision it ran under, so the split is auditable. Headline tables come only from frozen-tag runs over the development set of 20 oracle-certified tasks. A held-out set mined from etcd, caddy, and hugo is extracted and certified only after the tag, with the frozen engine; a certification failure there is reported as a protocol ceiling, not fixed.

## 5.2 Invalid intermediate states

The mechanism claim is that validation before disk changes what a repository passes through, not just where an episode ends. A watcher samples each worktree during the agent’s editing window: when the mtime fingerprint of the Go files changes and then holds still for one interval, the settled state gets a `go build ./...` verdict. The oracle runs as the control and measures the sampler itself. Raw and advisory arms pass through whatever the model writes; the semantic arm cannot land a declaration that does not typecheck, so its measured rate is the write window itself: a mutation touching dozens of call sites writes files sequentially, and a sample can catch the half-written state. Replaying every accepted mutation from the flagged episodes shows each settled state compiles; no host tool wrote Go in any of them.

# 6 Results

Pooled over the frozen grid, semantic passed 173/340 episodes; raw passed 78/340. The per-task table tells the sharper story: raw wins only where grep suffices, and goes to zero on the tasks that need whole-workspace consistency.

Two readings stand out. First, the arm ordering (Figure 1) never inverts across models: symbol addressing helps, and mandatory validation helps again on top of it. Second, the protocol’s advantage is not uniform. It is widest for the model that edits text worst: a strong instruction-follower that is nonetheless a poor freeform tool user passes almost nothing raw yet recovers to a majority through the protocol, because the protocol removes exactly the failure it is prone to. The invalid-intermediate rate (Figure 2) is the mechanism: both text arms leave the workspace uncompileable most of the time mid-edit, while the semantic arm holds it valid throughout.

# 7 Limitations

One language. Contamination is held constant across arms but inflates absolute rates. One serving stack per model, small k, and the infrastructure is author-run. TODO: expand honestly.

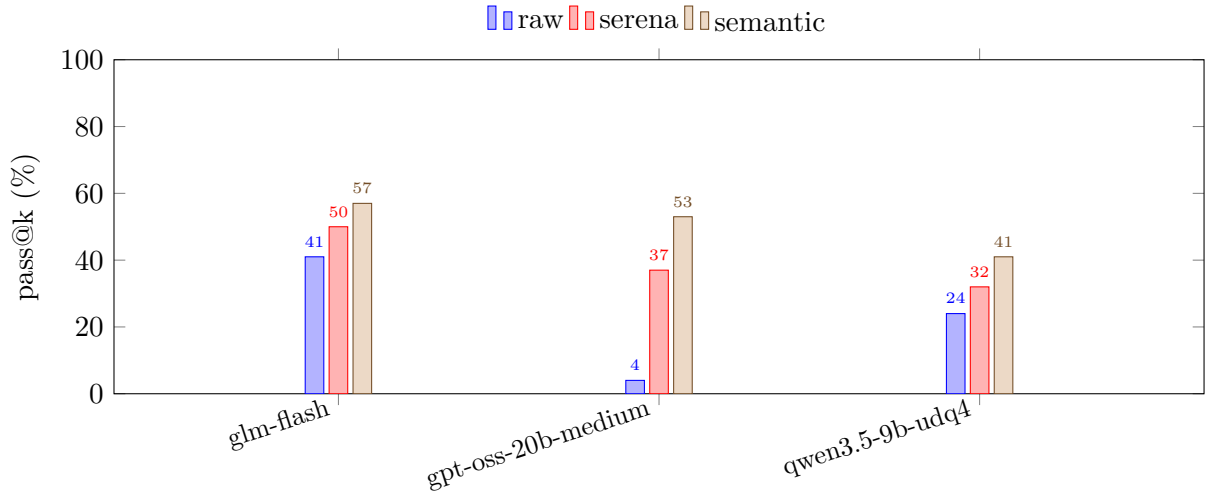


Figure 1: Pass@k by arm, per model. The ordering  $\text{semantic} \geq \text{serena} \geq \text{raw}$  holds for every model in the grid. The lift is largest where the model is a weak freeform tool-caller (gpt-oss-20b-medium: 4% raw, 53% semantic). Generated from committed episode evidence.

## 8 Availability

Code (MIT), all episode evidence, and analysis tooling: <https://github.com/guygrigsby/agent-go>.

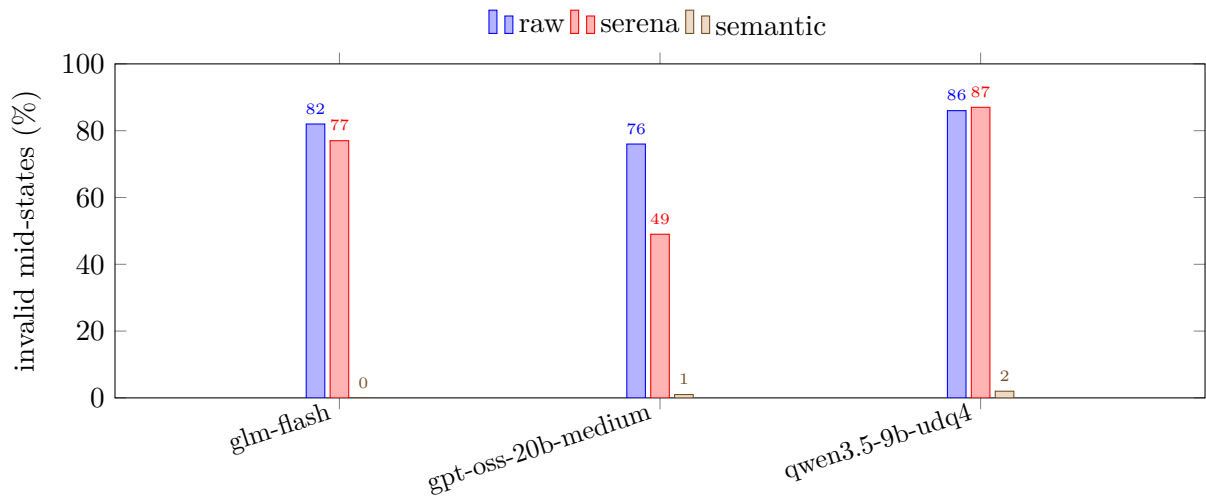


Figure 2: Fraction of sampled on-disk states that fail `go build` during the editing window. The text-editing arms churn through broken states; the semantic arm cannot land a declaration that does not typecheck, so its residual is only the multi-file write window. Generated from the same evidence.

| profile   | task              | mode     | pass@k | median green (s) | capped | resends |
|-----------|-------------------|----------|--------|------------------|--------|---------|
| glm-flash | boundary_06d69f69 | raw      | 6/6    | 244              | 0      | 0       |
| glm-flash | boundary_06d69f69 | semantic | 6/6    | 250              | 0      | 2       |
| glm-flash | boundary_06d69f69 | serena   | 6/6    | 178              | 0      | 0       |
| glm-flash | boundary_35b91c66 | raw      | 2/6    | 544              | 4      | 0       |
| glm-flash | boundary_35b91c66 | semantic | 3/6    | 379              | 3      | 12      |
| glm-flash | boundary_35b91c66 | serena   | 5/6    | 247              | 0      | 0       |
| glm-flash | boundary_57ffc718 | raw      | 0/6    | –                | 6      | 0       |
| glm-flash | boundary_57ffc718 | semantic | 2/6    | 469              | 4      | 29      |
| glm-flash | boundary_57ffc718 | serena   | 0/6    | –                | 6      | 0       |
| glm-flash | boundary_687dd1bd | raw      | 0/6    | –                | 3      | 0       |
| glm-flash | boundary_687dd1bd | semantic | 0/6    | –                | 6      | 72      |
| glm-flash | boundary_687dd1bd | serena   | 0/6    | –                | 5      | 0       |
| glm-flash | boundary_9237d6f7 | raw      | 1/6    | 554              | 2      | 0       |
| glm-flash | boundary_9237d6f7 | semantic | 3/6    | 401              | 2      | 98      |
| glm-flash | boundary_9237d6f7 | serena   | 5/6    | 175              | 0      | 0       |
| glm-flash | boundary_9354a4eb | raw      | 0/6    | –                | 6      | 0       |
| glm-flash | boundary_9354a4eb | semantic | 0/6    | –                | 6      | 11      |
| glm-flash | boundary_9354a4eb | serena   | 0/6    | –                | 6      | 0       |
| glm-flash | boundary_9f2c83f3 | raw      | 2/6    | 671              | 5      | 0       |
| glm-flash | boundary_9f2c83f3 | semantic | 5/6    | 407              | 0      | 6       |
| glm-flash | boundary_9f2c83f3 | serena   | 3/6    | 675              | 0      | 0       |
| glm-flash | boundary_b26814a3 | raw      | 1/6    | 720              | 5      | 0       |
| glm-flash | boundary_b26814a3 | semantic | 4/6    | 669              | 4      | 30      |
| glm-flash | boundary_b26814a3 | serena   | 6/6    | 406              | 0      | 0       |
| glm-flash | boundary_b4b95e0f | raw      | 0/6    | –                | 6      | 0       |
| glm-flash | boundary_b4b95e0f | semantic | 0/6    | –                | 6      | 7       |
| glm-flash | boundary_b4b95e0f | serena   | 0/6    | –                | 6      | 0       |
| glm-flash | boundary_bf1486f7 | raw      | 0/6    | –                | 6      | 0       |
| glm-flash | boundary_bf1486f7 | semantic | 4/6    | 657              | 4      | 8       |
| glm-flash | boundary_bf1486f7 | serena   | 0/6    | –                | 6      | 0       |
| glm-flash | boundary_c0532428 | raw      | 6/6    | 453              | 1      | 0       |
| glm-flash | boundary_c0532428 | semantic | 6/6    | 334              | 0      | 13      |
| glm-flash | boundary_c0532428 | serena   | 6/6    | 138              | 0      | 0       |
| glm-flash | boundary_caf19f86 | raw      | 6/6    | 715              | 3      | 0       |
| glm-flash | boundary_caf19f86 | semantic | 6/6    | 325              | 0      | 6       |
| glm-flash | boundary_caf19f86 | serena   | 6/6    | 171              | 0      | 0       |
| glm-flash | boundary_dd2c3807 | raw      | 1/6    | 900              | 3      | 0       |
| glm-flash | boundary_dd2c3807 | semantic | 3/6    | 720              | 6      | 21      |
| glm-flash | boundary_dd2c3807 | serena   | 1/6    | 741              | 4      | 0       |
| glm-flash | boundary_eb61ac63 | raw      | 0/6    | –                | 6      | 0       |
| glm-flash | boundary_eb61ac63 | semantic | 4/6    | 720              | 6      | 21      |
| glm-flash | boundary_eb61ac63 | serena   | 0/6    | –                | 6      | 0       |
| glm-flash | boundary_fb1ea92a | raw      | 6/6    | 431              | 0      | 0       |
| glm-flash | boundary_fb1ea92a | semantic | 0/6    | –                | 6      | 81      |
| glm-flash | boundary_fb1ea92a | serena   | 2/6    | 323              | 2      | 0       |
| glm-flash | traefik_51491463  | raw      | 5/6    | 272              | 0      | 0       |
| glm-flash | traefik_51491463  | semantic | 6/6    | 268              | 0      | 3       |
| glm-flash | traefik_51491463  | serena   | 5/6    | 128              | 0      | 0       |
| glm-flash | vault_7103bc2c    | raw      | 0/6    | –                | 3      | 0       |
| glm-flash | vault_7103bc2c    | semantic | 0/6    | –                | 6      | 20      |
| glm-flash | vault_7103bc2c    | serena   | 0/6    | –                | 5      | 0       |
| glm-flash | vault_95b24f55    | raw      | 6/6    | 340              | 0      | 0       |
| glm-flash | vault_95b24f55    | semantic | 6/6    | 324              | 0      | 4       |
| glm-flash | vault_95b24f55    | serena   | 6/6    | 248              | 0      | 0       |
| glm-flash | vault_caec65a7    | raw      | 1/6    | 720              | 6      | 0       |
| glm-flash | vault_caec65a7    | semantic | 6/6    | 425              | 0      | 0       |
| glm-flash | vault_caec65a7    | serena   | 5/6    | 456              | 2      | 0       |